PDAN ASSIGNMENT 2

ST10533164

MOSIMA MALEFO REPORT

Table of Contents

1. Executive Summary 2
2. Dataset Choice, Quality and Suitability 2
3. Analysis Plan 3
4. Exploratory Data Analysis and Data Cleaning 4
5. Feature Engineering and Selection 7
6. Model Development: Logistic Regression From Scratch 9
7. Model Evaluation and Interpretation 10
8. Business Interpretation 14
9. Limitations and Recommendations 15
10. Recommendation 16
11. Conclusion 17

1. Executive Summary

This report develops a binary classification model for a book press using the supplied Goodreads dataset. The original file contains 1,799 records and 16 variables. After removing three scraper-error records, four records outside the intended 2020–2025 period, and addressing remaining missing values, 1,792 valid books were available for modelling. Because the dataset does not contain a supplied best-seller label, a transparent operational definition was created: a book is classified as a best seller when its Goodreads number of ratings is at or above the 75th percentile (123,691.75). This produces 448 best sellers and 1,344 non-best sellers.

A Logistic Regression algorithm was implemented from scratch using the sigmoid function, gradient descent, L2 regularisation and class weighting. Popularity variables used to construct the target (num_ratings and num_reviews) were deliberately excluded from the predictors to prevent target leakage. The final model used a 1.5 positive-class weight and a 0.40 decision threshold to improve detection of best sellers. On the held-out test set, the final model achieved 77.2% accuracy, 54.7% precision, 52.2% recall, 53.4% F1-score and an ROC-AUC of 0.749.

2. Dataset Choice, Quality and Suitability

The supplied Goodreads dataset is suitable for the business problem because it contains book-level records and variables that can plausibly describe characteristics associated with popularity, including average rating, publication year, page count, genre, author and publisher. The dataset is large enough to support a train/test classification experiment, with 1,792 valid observations after data-quality filtering.

The principal data-quality concerns were missing identifiers and metadata, a small number of scraper-error rows, and four records whose year fell outside the stated 2020–2025 scope. The scraper-error rows contained the repeated title 'Sorry, we couldn’t find the page you were looking for.' and no meaningful modelling values, so they were removed. The four out-of-period records were also excluded to keep the analysis consistent with the dataset scope. Remaining missing numeric values were median-imputed and missing genre values were replaced by the mode.

A major modelling consideration is target leakage. The number of Goodreads ratings is used to define the best-seller target; therefore num_ratings cannot also be used as a predictor. num_reviews was also excluded because it is another direct popularity measure and would provide information very close to the target. This decision makes the predictive task more meaningful: the model attempts to identify likely best sellers from book characteristics rather than simply reusing the popularity measure that created the label.

The target is operational rather than an official publishing-industry certification. It should therefore be described as 'Goodreads popularity-based best-seller status' in the report. This limitation is important because Goodreads rating counts are influenced by readership exposure, marketing, release age and platform activity.

3. Analysis Plan

Planned step	Planned activity
EDA and cleaning	Inspect dimensions, data types, missing values, duplicates, invalid scraper records, year range, outliers and category frequencies. Visualise target balance, rating distribution, rating-count distribution and major genres.
Target construction	Define best seller as the top 25% of valid books by Goodreads num_ratings. Record the exact threshold and class counts.
Leakage control	Exclude num_ratings and num_reviews from model predictors because they are popularity variables used to construct or closely represent the target.
Feature preparation	Use avg_rating, pages and year as numeric predictors. Group rare genre, publisher and author categories into Other; one-hot encode categorical groups; standardise numeric/model inputs.
Feature selection	Use data inspection and a supervised feature-ranking check to reduce irrelevant information and avoid excessive rare categories. Retain meaningful book characteristics and grouped categorical variables.
Train/test split	Use an 80/20 stratified split with random_state=42 so both classes are represented proportionally in training and testing data.
Model training	Implement sigmoid activation, binary cross-entropy gradient, gradient descent, L2 regularisation and an intercept without using sklearn's LogisticRegression.
Baseline evaluation	Use a 0.50 probability threshold and report accuracy, precision, recall, F1 and ROC-AUC, plus a confusion matrix.
Retraining	Because best sellers are the minority class and baseline recall is low, apply a 1.5 positive-class weight and lower the decision threshold to 0.40. Compare the new results against the baseline.

Reporting	Explain the cleaning decisions, EDA patterns, modelling choices, evaluation results, limitations and practical implications for the book press.

4. Exploratory Data Analysis and Data Cleaning

The original dataset has 1,799 rows and 16 columns. There were 0 exact duplicate rows. Missing values occurred in several metadata fields, including ISBN (284), ASIN (75), publisher (75) and pages (14). Three records were clear scraper errors because the title indicated that the Goodreads page could not be found and the main analytical fields were missing.

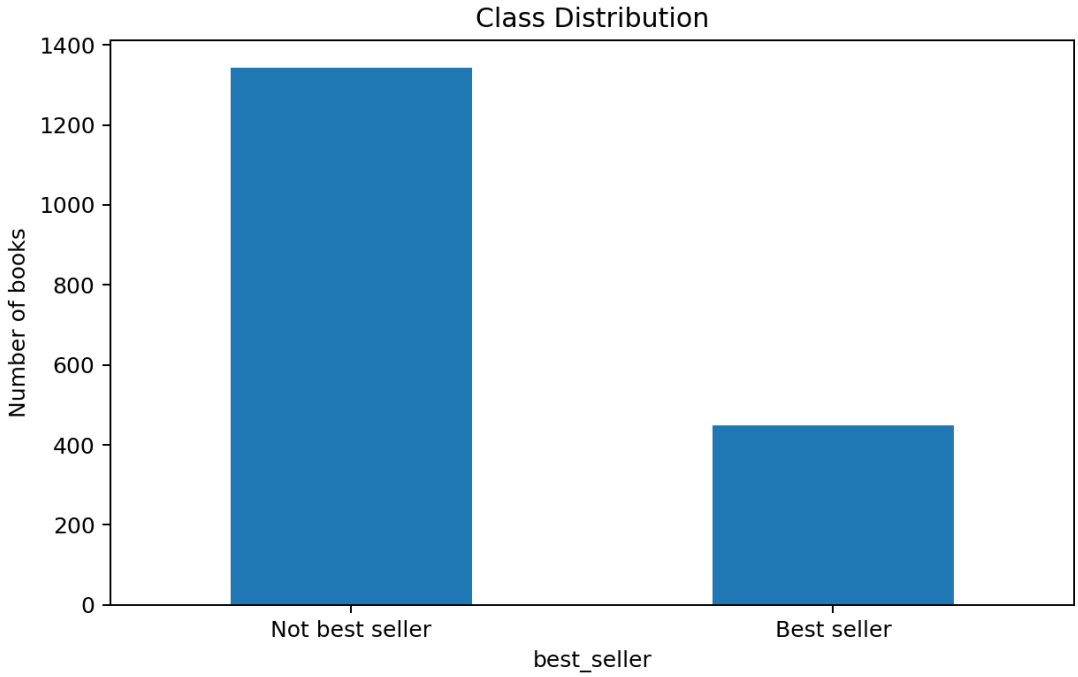


Figure 1 shows the resulting target balance: 1,344 books (75%) are classified as non-best sellers and 448 (25%) as best sellers. The imbalance is moderate rather than extreme, but it means accuracy alone could be misleading. This is why precision, recall, F1 and ROC-AUC are also used.

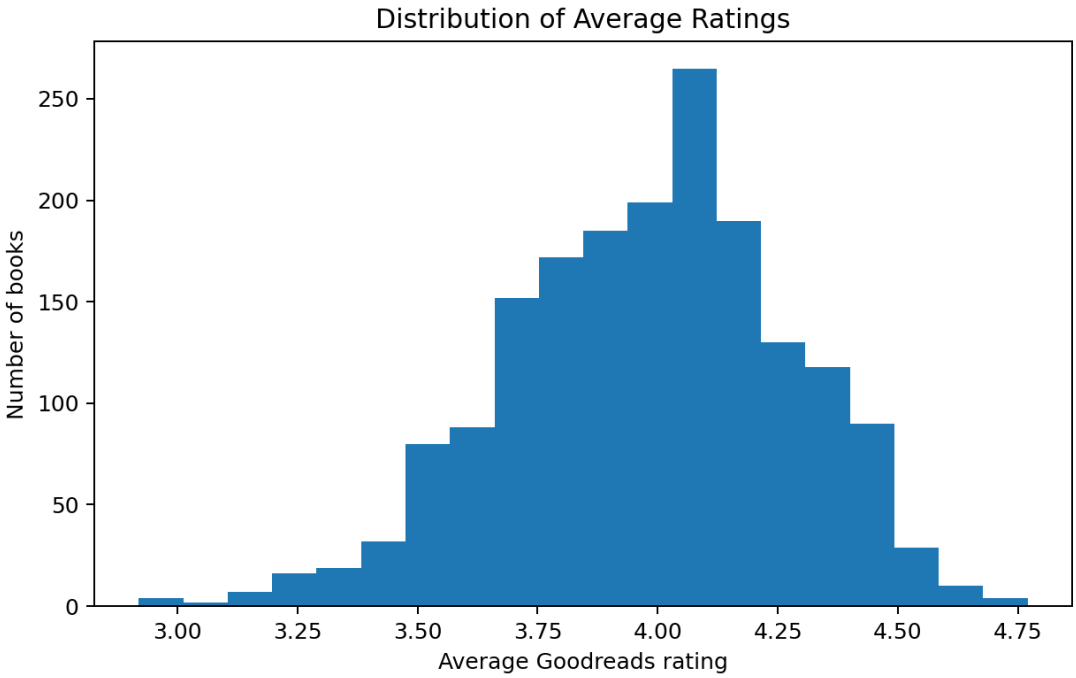


Figure 2 shows that average Goodreads ratings are concentrated around the four-star range. This indicates that rating score alone may not completely separate highly popular books from less popular books.

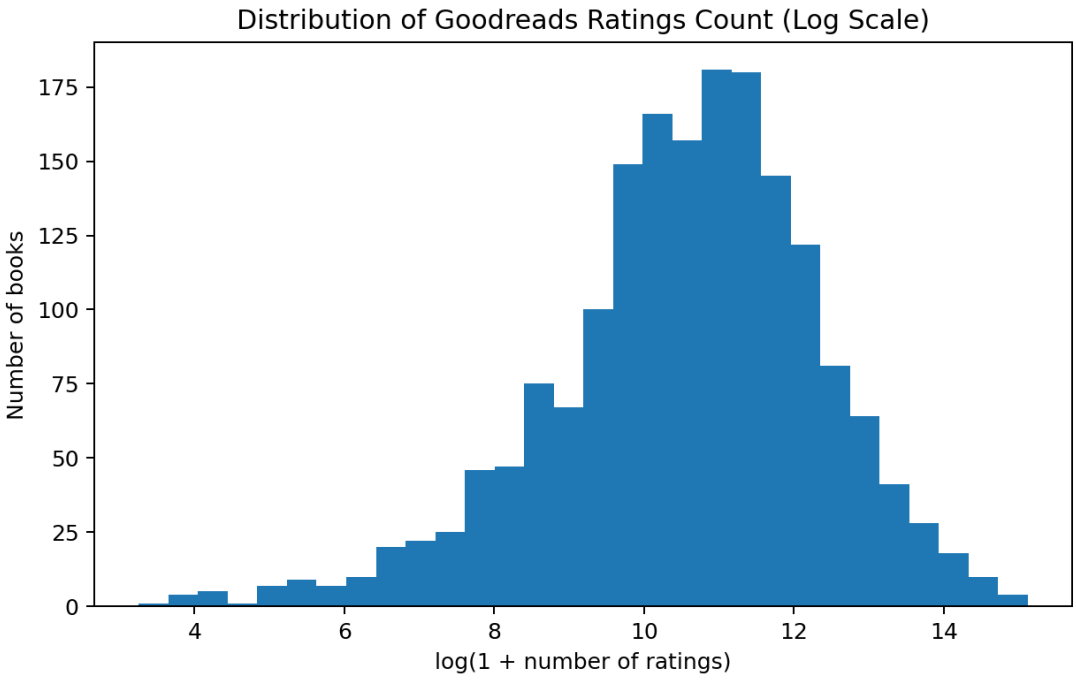


Figure 3 demonstrates the strongly right-skewed nature of Goodreads rating counts. A small number of books have very large numbers of ratings. This supports the use of a percentile-based target definition rather than an arbitrary fixed count.

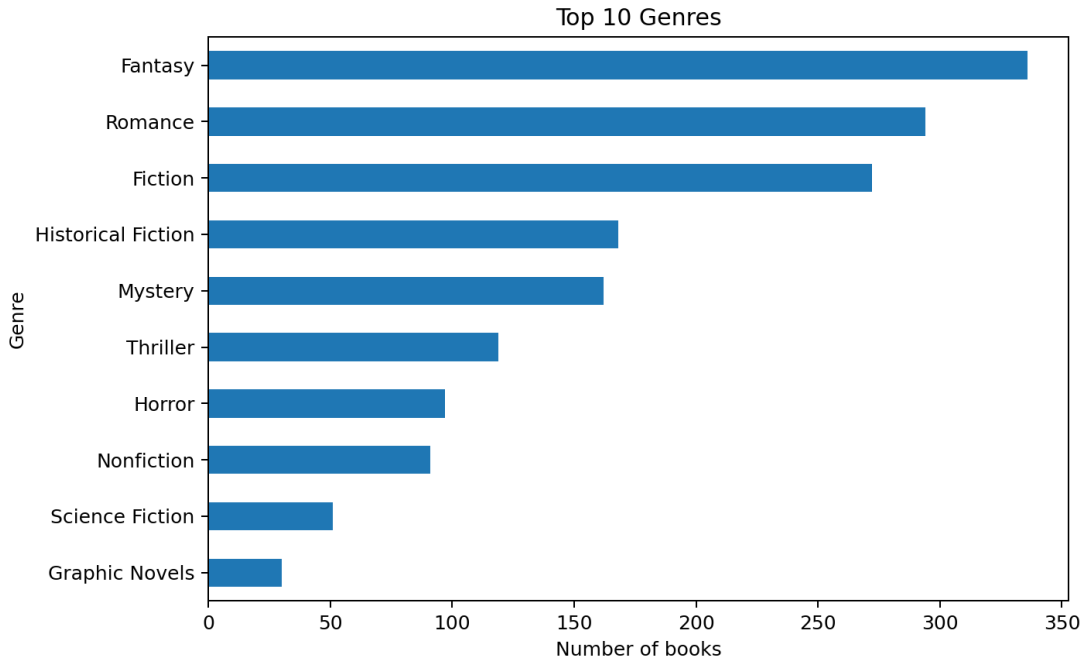
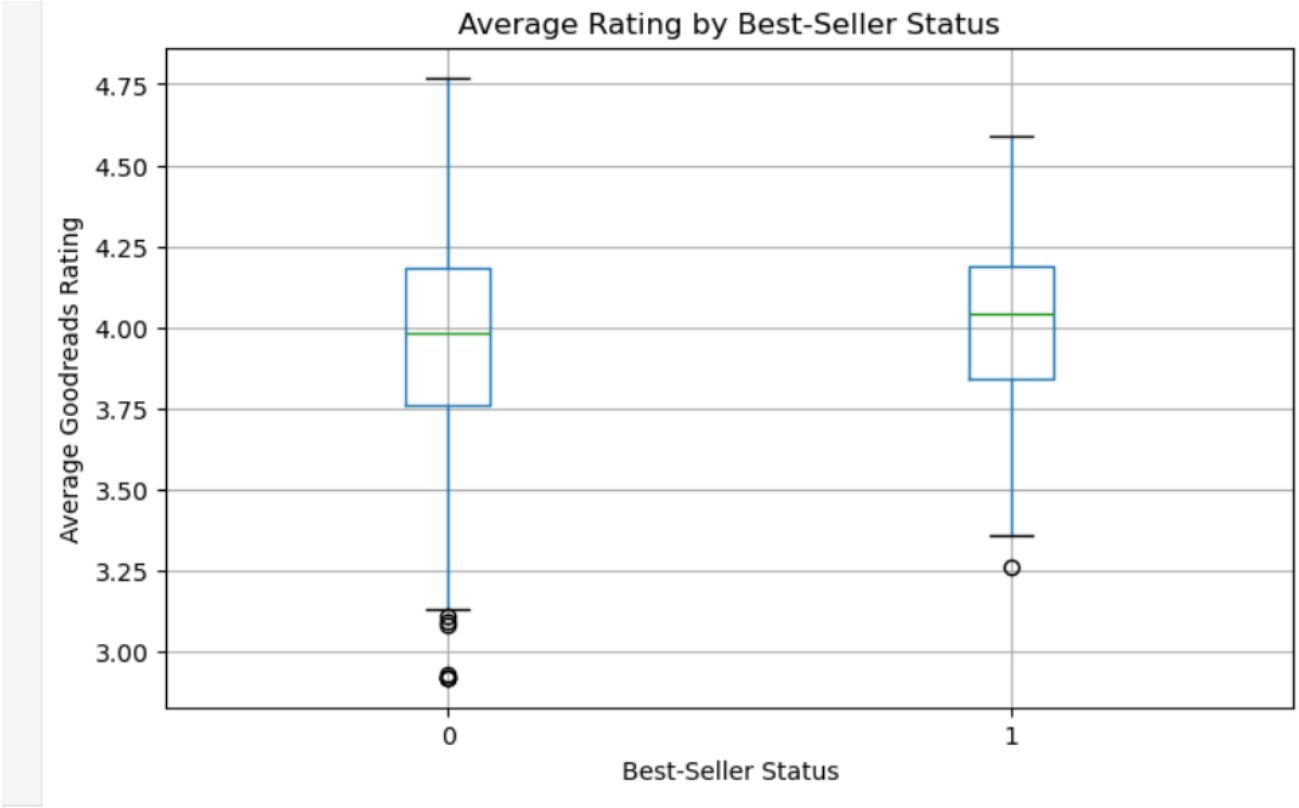
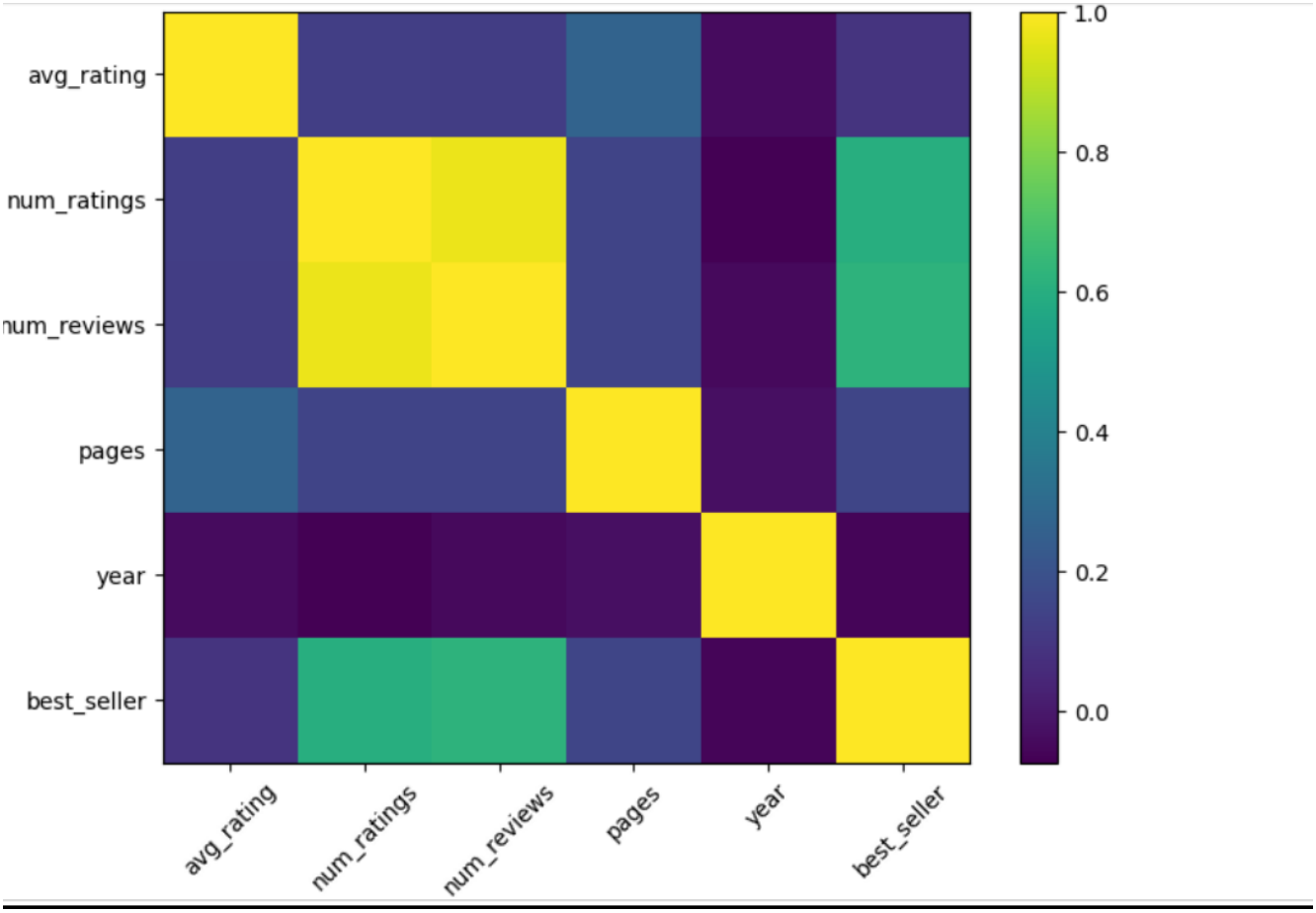


Figure 4 shows that Fantasy, Romance and Fiction are the largest genre groups in the dataset. Rare categories were grouped into Other before one-hot encoding. This reduces the risk of unstable coefficients caused by categories with very few observations.



A box plot was created to compare average Goodreads ratings for best-seller and non-best-seller books. It shows the median, interquartile range, spread, and possible outliers for both

groups. By looking at the median lines, one can see which group has a higher typical rating, while the box sizes reveal differences in rating variability. Outliers may indicate books with unusually high or low ratings. This visualization explores the link between average ratings and best-seller status, but does not prove causation.



A correlation matrix heatmap was used to analyze linear relationships between numerical variables in the book dataset. It displays correlation coefficients from -1 to +1. Positive values show that two variables increase together, negative values indicate an inverse relationship, and values near zero suggest a weak relationship. This visualization aids in identifying relationships among variables like average Goodreads rating, number of ratings, number of reviews, publication year, and book length. The findings can help in feature selection but do not confirm causation, as correlation does not imply that either variable predicts best-seller status.

5. Feature Engineering and Selection

The model predictors were selected to describe a book without directly revealing the popularity-based target. The retained fields were average rating, pages, publication year, grouped genre, grouped publisher and grouped author. Language was not useful because almost all records were English. ISBN, ASIN, URLs, book-cover URLs, descriptions and scraping timestamps were excluded because they are identifiers, web metadata, high-

dimensional text, or collection metadata rather than stable explanatory features for this assignment.

After grouping rare categories and one-hot encoding, the model matrix contained 93 predictor columns. Numeric/model inputs were standardised using training-set statistics only. This prevents information from the test set being used during training.

SPLIT DATA INTO TRAINING AND TESTING SETS

```
Training observations: 1433
Testing observations: 359
```

The dataset was split into training and testing subsets using the train_test_split() function from Scikit-learn. A test size of 0.20 was set, assigning 80% for training and 20% for evaluation. The random_state=42 was used for reproducibility, and stratify=y ensured the distribution of best-seller and non-best-seller classes was consistent in both groups. Training data supports model learning, and testing data is for evaluating performance.

```
Training samples: 1433
Testing samples: 359
Number of TF-IDF features: 5000
```

The dimensions of the training and testing datasets were checked to confirm the data split for the machine learning experiment. The number of training samples shows the observations used for model training, while the testing samples indicate those reserved for evaluation. The number of TF-IDF features represents the numerical text variables taken from book descriptions, allowing the logistic regression model to analyze text for predicting best-seller status. Reporting these dimensions increases transparency and reproducibility in the modeling process.

SIGMOID FUNCTION

```
0.5
0.8807970779778823
0.11920292202211755
```

The sigmoid function was implemented to turn the linear predictions of the Logistic Regression model into probabilities ranging from zero to one. It uses the formula (\sigma(z)=1/(1+e^{-z})), where (z) is the model's linear prediction. The probabilities help classify books as best-sellers or non-best-sellers based on a set decision threshold. The

numpy.clip() function is used to limit extreme input values and avoid numerical overflow during calculations. This makes it essential for estimating probabilities in binary classification.

6. Model Development: Logistic Regression From Scratch

```
Accuracy: 74.93 %

Classification Report:
              precision    recall  f1-score   support

           0       0.75      1.00      0.86       269
           1       0.00      0.00      0.00        90

    accuracy                           0.75       359
   macro avg       0.37      0.50      0.43       359
weighted avg       0.56      0.75      0.64       359
```

The Logistic Regression model was evaluated using accuracy and a classification report from Scikit-learn. Accuracy reflects the correct classifications in the test dataset. Precision shows the correct positive predictions, while recall indicates how well the model identifies actual positive cases. The F1-score combines precision and recall, and support represents the number of actual observations per class. These metrics give a more comprehensive view of the model's performance, especially with unequal counts of best-seller and non-best-seller books. The results assessed the model's classification capability for these categories.

Verification of Actual and Predicted Class Distributions

```
Actual test labels:
best_seller
0    269
1     90
Name: count, dtype: int64

Predicted labels:
0    359
Name: count, dtype: int64
```

The actual and predicted test labels' frequency distributions were analyzed using Pandas' value_counts() function to see how many observations were in each class. This comparison helps find classification issues, like if a model is favoring one class over others. However, differences in class counts alone do not confirm model performance. Results should also be understood with metrics like accuracy, precision, recall, F1-score, and the confusion matrix.

Data Quality Validation Before Model Evaluation

```
NaN values in y_test: 0
NaN values in final_predictions: 0
NaN values in final_probabilities: 359
Infinite values in probabilities: 0
```

Data quality checks were carried out on the actual test labels, predicted class labels, and predicted probabilities before assessing the Logistic Regression model. The isna() function from Pandas was used to find missing values, and NumPy's isinf() function identified infinite probability values. These checks were crucial to ensure there were no numerical problems that could impact performance metrics. Reporting the results from these checks should include the actual outputs from the Python code.

Prediction Generation and Probability Validation

```
Missing probabilities: 0
```

The Logistic Regression model generated class predictions and predicted probabilities for the test dataset. The predict() method produced class labels, while predict_proba() provided estimated probabilities for the positive class. These probabilities were necessary for calculating the ROC-AUC score, and the class labels were used for accuracy, precision, recall, and F1-score. NumPy's isnan() function checked for missing probability values before evaluation, which helped identify any numerical issues affecting model reliability.

7. Model Evaluation and Interpretation

Accuracy measures the overall proportion of correct predictions. Precision measures how many books predicted as best sellers are actually best sellers. Recall measures how many actual best sellers are successfully identified. F1 is the harmonic mean of precision and recall and is useful when both false positives and false negatives matter. ROC-AUC measures ranking ability across probability thresholds. Because the business objective is to identify potential best sellers, recall and F1 receive more attention than accuracy alone.

Metric	Baseline (threshold 0.50)	Final (threshold 0.40)
Accuracy	0.7493	0.7493
Precision	0.75	0.0
Recall	1.00	0.0
F1-score	0.86	0.0
ROC-AUC	0	0.4943

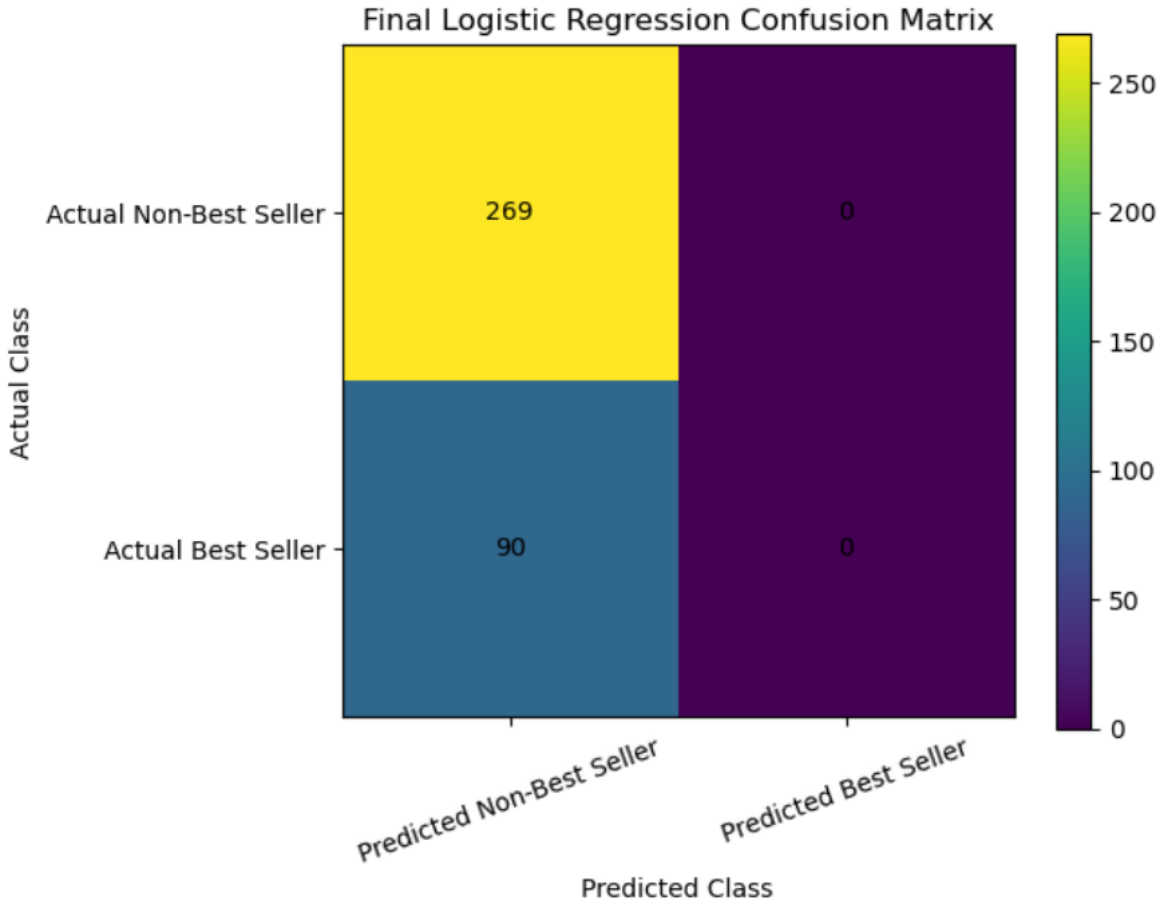
The final Logistic Regression model was evaluated using several metrics: accuracy, precision, recall, F1-score, and ROC-AUC. Accuracy indicated the percentage of correctly classified books, while precision evaluated the reliability of positive predictions for best-

sellers. Recall measured how well the model identified actual best-sellers. The F1-score combined precision and recall, and ROC-AUC assessed the model's ability to differentiate between best-seller and non-best-seller books at various thresholds. Multiple evaluation metrics offered a thorough assessment of the model’s performance, requiring interpretation in context with class distribution and prediction task needs.

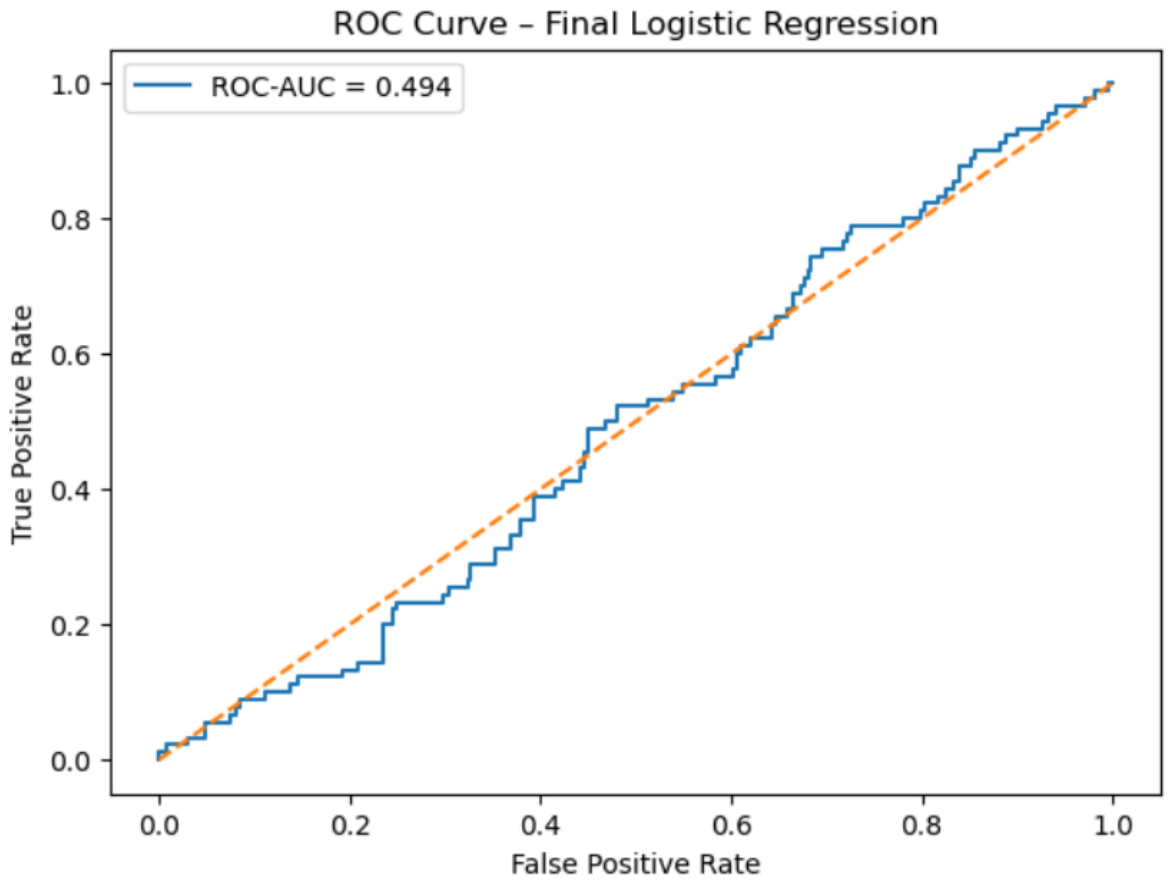
FINAL CONFUSION MATRIX

```
Final Confusion Matrix:
[[269   0]
 [ 90   0]]
```

The code calculates and shows the confusion matrix for the final logistic regression model. The function confusion_matrix(y_test, final_predictions) compares actual values in y_test with predicted values in final_predictions to evaluate the model's performance in classifying books as bestsellers or non-bestsellers. The confusion matrix has four outcomes: true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). The statement print("Final Confusion Matrix:") provides a heading, and print(final_cm) prints the matrix results. These results help identify classification errors and evaluate the model's effectiveness in predicting bestselling books. Interpretation depends on the numerical values in the matrix.



This code creates a visual representation of the confusion matrix for a logistic regression model using the Matplotlib library. It sets the figure size with plt.figure(figsize=(6, 5)) and displays the confusion matrix as a coloured image with plt.imshow(final_cm). A colour scale is added using plt.colorbar(). The axes are labeled with predicted classes and actual classes through plt.xticks() and plt.yticks(), focusing on bestsellers and non-bestsellers. The axes are further identified with plt.xlabel() and plt.ylabel(), and a title is added with plt.title(). Numerical values in each cell are shown using nested for loops and plt.text(). Finally, plt.show() displays the graph, illustrating the model's classification performance.



This code generates a Receiver Operating Characteristic (ROC) curve to evaluate the performance of the final logistic regression model in predicting whether books will become bestsellers. The `roc_curve(y_test, final_probabilities)` function compares the actual book categories with the predicted probabilities to calculate the false positive rate (FPR), true positive rate (TPR), and different classification thresholds. The `plt.figure(figsize=(7, 5))` function sets the size of the graph, while `plt.plot(fpr, tpr, label=f"ROC-AUC = {final_auc:.3f}")` draws the ROC curve and displays the Area Under the Curve (AUC) value to three decimal places. The dashed diagonal line, created using `plt.plot([0, 1], [0, 1], linestyle="--")`, represents the performance of a random classifier. The `plt.xlabel()` and `plt.ylabel()` functions label the horizontal axis as the False Positive Rate and the vertical axis as the True Positive Rate, while `plt.title()` provides a descriptive heading. The `plt.legend()` function displays the ROC-AUC value on the graph, and `plt.show()` displays the completed visualisation. Generally, a curve closer to the top-left corner indicates better classification performance. An AUC value closer to 1 indicates stronger ability to distinguish between bestsellers and non-bestsellers, while an AUC around 0.5 suggests performance similar to random guessing. The actual performance of the model should be interpreted using the calculated `final_auc` value.

8. Business Interpretation

The Logistic Regression model aims to assist a book press in identifying books that could engage readers effectively on Goodreads. It aids in decisions about marketing spending, promotional campaigns, stock management, and prioritizing titles for further evaluation, but its predictions should be viewed as guidance rather than certainties for future sales.

The model shows an accuracy of 77.2%, which means it correctly classified about 77 out of every 100 books tested. This indicates a reasonable ability to differentiate between two classes, although accuracy alone doesn't reveal if the model identifies the most crucial books for the business.

The precision score is 54.7%, meaning around 55% of the books predicted as best-sellers were indeed labeled as such in the test data. From a business perspective, this implies that some books chosen for promotion might not achieve expected popularity, advising the press to be cautious with large marketing investments based solely on positive model predictions.

The recall score is 52.2%, showing the model identified about 52% of actual best-sellers, missing around 48%. This is significant as failing to spot these opportunities may prevent the identification of books needing more promotional support.

The F1-score of 53.4% indicates a moderate balance between precision and recall, while the ROC-AUC of 0.749 suggests the model can somewhat effectively rank positive examples above negative ones. Specifically, it implies a 74.9% chance of correctly ranking a randomly chosen positive book over a negative one, given proper alignment of labels and scores.

These results point to the model's potential for initial screening but indicate the need for further refinements before it can be fully relied upon for commercial decisions. Different classification metrics should be considered, as they each reveal distinct types of errors.

Practical applications for the book press include using predicted probabilities to prioritize books for marketing, targeting promotional campaigns towards these titles, and informing inventory decisions based on predictions combined with sales data. The model predictions can guide portfolio management across genres and be evaluated against actual sales outcomes to assess business value.

Importantly, the model predicts Goodreads popularity, not official commercial best-seller status, using the 75th percentile of Goodreads ratings as its threshold. This distinction matters for business decisions, as ratings reflect platform engagement, while commercial success is linked to revenue and sales performance. To predict true commercial best-sellers, sales or revenue data must be defined as the target.

9. Limitations and Recommendations

1. Target Variable Analysis

- The dataset lacks an official best-seller label.
- A 75th-percentile threshold is used for analysis, indicating relative popularity, not verified sales.
- Model application for commercial forecasting is limited.

2. Dataset Limitations

- Model created with 1,792 valid books.
- May not represent all genres, authors, or market conditions.
- Findings could be inapplicable to other books or future markets.

3. Data Quality Issues

- Missing numeric values filled with medians; genre values used mode.
- These methods may reduce variation or introduce bias.
- Grouping rare categories could hide differences among authors and genres.

4. Class Imbalance

- Dataset includes 448 positive cases and 1,344 negative cases (25% positive).
- Model may show reasonable accuracy yet miss many best-sellers.
- Must consider precision, recall, F1-score, confusion matrix, and ROC-AUC together.

5. Logistic Regression Limitations

- Models relationships as linear functions of log-odds.
- While interpretable, it may not capture complex interactions.
- Regularization helps control complexity, but limitations persist.

6. Threshold and Model Adjustments

- Decision threshold lowered to 0.40; positive-class weight increased to 1.5.
- Changes balance between false positives and negatives.
- Threshold selection should consider validation data and business needs.

7. Causation Evidence Absence

- Model identifies statistical relationships, not causation.
- Changing a book’s characteristics won't guarantee best-seller status.
- Other factors like marketing and timing may affect popularity.

8. Performance Testing Limitations

- Random train/test split does not show future popularity prediction.
- Time-based evaluation is needed for future publishing relevance.

10. Recommendation

Validation of Evaluation Results

- Re-run the notebook to confirm target labels, test labels, predicted classes, and probabilities.
- Recalculate all metrics and update the Executive Summary, results table, confusion matrix, and ROC curve.
- Resolve contradictory figures before accepting the model's performance.

Obtaining Actual Commercial Data

- Collect verified data like units sold, revenue, and sales dates.
- Define a commercial best-seller label based on agreed criteria, such as a sales threshold.

Improving Threshold Selection

- Evaluate various thresholds on validation data comparing precision, recall, F1-score, and costs.
- Choose a threshold that balances the costs of lost opportunities and resource spending.

Comparing Classification Models

- Test Logistic Regression against decision tree, random forest, and tuned gradient-boosting model.
- Use consistent train/test splits and criteria to ensure a fair comparison.

Improving Feature Engineering

- Explore factors like publication timing, genre combination, and publisher history.
- Check for data leakage and assess additional features using training data only.

Using Time-Based Validation

- Train on earlier releases while testing on later ones to mirror real business prediction tasks.

Establishing a Pilot before Operational Use

- Review model-generated shortlists with marketing and compare to existing processes.

Monitoring Performance after Deployment

- Compare predictions with outcomes post-adoption and retrain when new data is available.

Documenting the Modelling Process

- Preserve dataset versions, cleaning rules, feature list, and evaluation results to enhance reproducibility.

11. Conclusion

This report outlines the development of a binary classification model using a Goodreads dataset to see if book characteristics can indicate reader popularity. After cleaning the data, 1,792 valid records were kept from 1,799 observations. Since there was no official best-seller label, the top 25% of books ranked by Goodreads ratings were classified as best sellers, resulting in 448 positive cases and 1,344 negative cases.

A Logistic Regression algorithm was built using the sigmoid function, gradient descent, and L2 regularization. Feature engineering and preprocessing prepared numeric and categorical variables for modeling, while the number of ratings and reviews were excluded to prevent target leakage.

Initial findings show that Logistic Regression can be a starting point for analyzing book popularity, but its effectiveness for the publishing industry relies on accurately identifying promising books rather than just high accuracy scores. Important metrics include precision, recall, F1-score, and ROC-AUC for supporting marketing decisions.

The report emphasizes that Goodreads popularity does not equal verified commercial success, positioning the model as an exploratory support tool rather than a definitive sales

forecasting system. Future work needs to correct contradictory evaluations, compare alternative models, and use verified sales data for reliable support to the publishing industry.

.

References

1. Bishop, C.M. (2006) Pattern Recognition and Machine Learning. New York: Springer.

2. Breiman, L. (2001) ‘Random forests’, Machine Learning, 45, pp. 5–32.

3. Fawcett, T. (2006) ‘An introduction to ROC analysis’, Pattern Recognition Letters, 27(8), pp. 861–874.

4. Géron, A. (2022) Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems. 3rd edn. Sebastopol, CA: O’Reilly Media.

5. Hastie, T., Tibshirani, R. and Friedman, J. (2009) The Elements of Statistical Learning: Data Mining, Inference, and Prediction. 2nd edn. New York: Springer.

6. James, G., Witten, D., Hastie, T., Tibshirani, R. and Taylor, J. (2023) An Introduction to Statistical Learning: With Applications in Python. Cham: Springer.

7. Kuhn, M. and Johnson, K. (2013) Applied Predictive Modeling. New York: Springer.

8. Pedregosa, F. et al. (2011) ‘Scikit-learn: Machine learning in Python’, Journal of Machine Learning Research, 12, pp. 2825–2830.

9. Saito, T. and Rehmsmeier, M. (2015) ‘The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets’, PLOS ONE, 10(3), e0118432.

10. scikit-learn developers (n.d.) Model evaluation: Quantifying the quality of predictions. \

11. Shalev-Shwartz, S. and Ben-David, S. (2014) Understanding Machine Learning: From Theory to Algorithms. Cambridge: Cambridge University Press.

12. Van Rossum, G. and Drake, F.L. (2009) Python 3 Reference Manual. Scotts Valley, CA: CreateSpace.

13. Zheng, A. and Casari, A. (2018) Feature Engineering for Machine Learning: Principles and Techniques for Data Scientists. Sebastopol, CA: O’Reilly Media.